

Design note: the unified VCE module (src/vce.c)

Status: approved design, implementation begins v1.6.33. Owner: Mico Mrkaic. Drafted from the code audit of 2026-07-31.

Problem

tea computes robust/clustered standard errors in FOUR independent implementations: regress (HC1 + CR1 in regress.c), the GLM family logit/probit/poisson (score sandwich in do_glm_binary), ivregress (its own 2SLS sandwich), and xtreg (within-model sandwich). Audit findings:

1. Drift risk is structural: four copies of the same mathematics, each with its own finite-sample adjustment code, none sharing tests.
2. One semantic BUG: xtreg, fe robust computes plain HC1 on the within regression. Stata (since release 10) silently promotes robust to vce(cluster panelvar) for xtreg because HC1 is inconsistent under the within transformation (Stock & Watson, Econometrica 2008). tea currently prints a different — wrong — number than Stata for the same command line.
3. The surface syntax is accidentally consistent today (bare robust / cluster() and vce(robust) / vce(cluster v) both parse in the audited paths) but only because the parsing code was copied around; nothing enforces it.

The abstraction

Every sandwich VCE in scope is $V = c \cdot B \cdot M \cdot B$ where

- B (“bread”, $K \times K$) is estimator-specific and already computed by each estimator: $(X'X)^{-1}$ for OLS and the within model; $(\hat{X}'\hat{X})^{-1}$ for 2SLS; the inverse negative Hessian for ML.
- M (“meat”) is built from per-observation SCORE rows s_i ($1 \times K$): $s_i = e_i \cdot x_i$ (OLS/within), $e_i \cdot \hat{x}_i$ with e_i from the ACTUAL regressors (2SLS), the gradient contribution (ML). robust: $M = \sum_i s_i' s_i$ cluster: $M = \sum_g u_g' u_g$, $u_g = \sum_{i \in g} s_i$
- c is a finite-sample adjustment from the policy table below.

The module therefore needs exactly three things from an estimator: bread, the score matrix, and its row in the policy table. The plain (non-robust) VCE never touches the module — it stays estimator-owned ($\sigma^2 B$ for least squares, $-H^{-1}$ for ML).

Policy table (the Stata-compatibility core)

estimator	robust c	cluster c	test stat, df
regress	N/(N-k) (HC1)	$G/(G-1) \cdot (N-1)/(N-k)$ (CR1)	t, N-k / t, G-1
ivregress 2sls	N/(N-k)	$G/(G-1) \cdot (N-1)/(N-k)$	z (Stata uses z)
logit/probit/ poisson (ML)	N/(N-1)	$G/(G-1)$	z, unchanged
xtreg, fe	PROMOTED to cluster (panelvar) + note	$G/(G-1) \cdot (N-1)/(N-k)$, k = within slopes + intercept, absorbed FEs NOT counted	t, G-1; model F df (k_w, G-1)

Cells in this table are hypotheses to be pinned by the Stata verification run (tools/stata_check_vce.do) under the 6-digit contract; any cell Stata contradicts is corrected to Stata, and the table is the record.

API

```
typedef enum { VCE_DEFAULT, VCE_ROBUST, VCE_CLUSTER } VceKind;
typedef struct {
    VceKind kind;
    char    cluster_var[33];
} VceSpec;

/* one parser for the whole surface: bare robust / cluster(v)
 * and vce(robust|hcl|cluster v). Returns 0, or 198 with a
 * message already printed for malformed input. */
int vce_parse(const char *options, VceSpec *spec);

/* V = c · B M B, K×K malloc'd; scores col-major N×K; cid may be
 * NULL for robust. Omitted-column expansion is the caller's
 * concern (it owns the omitted map). */
double *vce_sandwich(const double *bread, const double *scores,
                    long N, int K, const long *cid, long G,
                    double c);

/* the shared table furniture: "Robust" column header text and
 * the "(Std. err. adjusted for G clusters in var)" note, so all
```

```

* estimators print identically. */
const char *vce_se_header(const VceSpec *s);
void          vce_print_cluster_note(const VceSpec *s, long G);

```

Migration plan

1. v1.6.33: vce.c lands. regress, glm, ivregress ported with NUMBERS UNCHANGED (their current adjustments already follow the table; porting is refactor, verified by the goldens not moving).
2. Same release: xtreg robust FIXED to promote to cluster(panelvar). This is a deliberate golden change wherever a test exercised the old wrong number, documented as the semantic bug it is.
3. Test 68: the estimator $\times$ {default, robust, cluster} matrix on grunfeld and nmes1988, locking cross-rig determinism.
4. tools/stata_check_vce.do: the same matrix for Stata at the IMF; its output pins the policy table to 6 digits.

Non-goals (deliberately)

HC2/HC3, two-way clustering, bootstrap, jackknife, and HAC (newey) are OUT of this release. The scores-based meat interface is chosen so HAC becomes “a different M from the same s_i” when the time-series tier arrives; nothing else will need to change.